IB MATHEMATICS SL • UNIT 3.01

Topic 3: Geometry, Trigonometry & Voronoi Diagrams

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

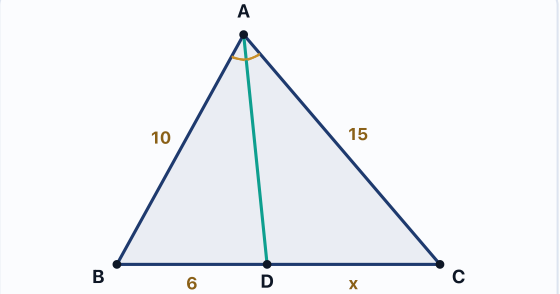
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 A	Q2 A	Q3 C	Q4 C	Q5 A
Q6 D	Q7 A	Q8 C	Q9 C	Q10 A
Q11 B	Q12 C	Q13 C	Q14 D	Q15 D
Q16 D	Q17 D	Q18 D	Q19 A	Q20 C
Q21 B	Q22 D	Q23 B	Q24 A	Q25 A

Topic 3: Geometry, Trigonometry & Voronoi Diagrams Practice Exercises

1 MCQ

In the given triangle ABC, AB = 10, AC = 15, and BD = 6. D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.



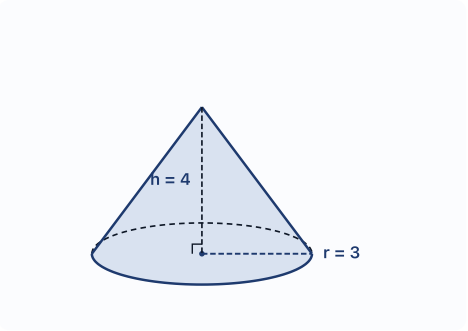
- **A** 9 ✓
- **B** 10
- **C** 15
- **D** 22.5

WORKED SOLUTION & VERIFICATION

By the Angle Bisector Theorem, the ratio of the sides adjacent to the angle bisector is equal to the ratio of the segments created by the bisector on the opposite side. Therefore, $\frac{AB}{AC} = \frac{BD}{DC}$.
Substituting the given values, we get $\frac{10}{15} = \frac{6}{DC}$.
Simplifying the left side, $\frac{2}{3} = \frac{6}{DC}$.
Cross-multiplying, $2 \times DC = 3 \times 6$.
$2 \times DC = 18$.
Dividing by 2, $DC = \frac{18}{2} = 9$. **[M1] [A1]**

2 MCQ

Volume of cone with radius 3, height 4:



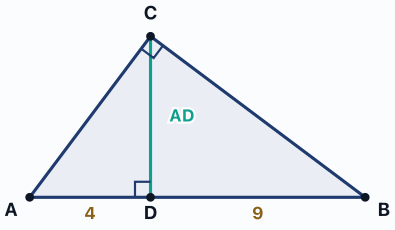
A 12π ✓	**B** 16π
C 36π	**D** 48π

WORKED SOLUTION & VERIFICATION

Applying Geometry & Trigonometry principles, $V = \frac{1}{3}\pi(9)(4) = 12\pi$. Therefore, the verified correct option is 12π. **[M1] [A1]**

3 MCQ

In the given right-angled triangle ABC, angle B is 90 degrees. Point D is on BC such that BD = 4 and DC = 9. If AD is the altitude to the hypotenuse AC, find the length of AD.



A 8	**B** 12
C 6 ✓	**D** 15

WORKED SOLUTION & VERIFICATION

In a right-angled triangle, the altitude to the hypotenuse creates two smaller triangles that are similar to the original triangle and to each other.

According to the Geometric Mean Theorem, the square of the altitude to the hypotenuse is equal to the product of the lengths of the two segments it divides the hypotenuse into.

In triangle ABC, AD is the altitude to the hypotenuse BC. Therefore, we have:

$AD^2 = BD \times DC$

Given that BD = 4 and DC = 9:

$AD^2 = 4 \times 9$

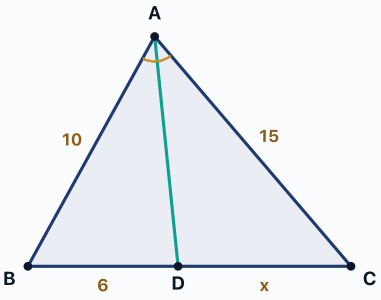
$AD^2 = 36$

$AD = \sqrt{36}$

$AD = 6$. **[M1] [A1]**

4 MCQ

In the triangle ABC, AB = 10, AC = 15, BD = 6, and D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.



A 12

B 7

C 9 ✓

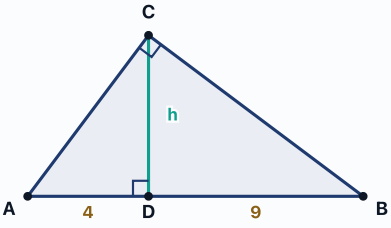
D 10

WORKED SOLUTION & VERIFICATION

According to the Angle Bisector Theorem, the ratio of the sides adjacent to the angle bisector is equal to the ratio of the segments it divides the opposite side into. Therefore, $\frac{AB}{AC} = \frac{BD}{DC}$. Substituting the given values, we have $\frac{10}{15} = \frac{6}{x}$, where x is the length of DC. Solving for x: $10x = 15 \times 6$ $10x = 90$ $x = \frac{90}{10}$ $x = 9$. **[M1] [A1]**

5 MCQ

In the diagram, triangle ABC has angle ABC equal to 90 degrees. Point D lies on AC such that BD is perpendicular to AC. If AD = 4 cm and DC = 9 cm, find the length of BD.



A 6 cm ✓

B 5 cm

C 7 cm

D 3 cm

WORKED SOLUTION & VERIFICATION

In a right-angled triangle, when an altitude is drawn from the right angle to the hypotenuse, it creates two smaller triangles that are similar to the original triangle and to each other. This leads to the geometric mean theorem. The length of the altitude (BD) is the geometric mean of the two segments it divides the hypotenuse into (AD and DC).

Therefore, $BD^2 = AD \times DC$.

Given AD = 4 cm and DC = 9 cm,

$BD^2 = 4 \times 9$

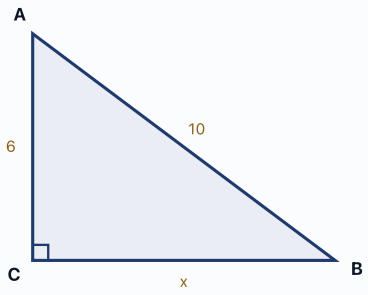
$BD^2 = 36$

$BD = \sqrt{36}$

$BD = 6$ cm. **[M1] [A1]**

6 MCQ

In the right-angled triangle ABC shown below, the length of side AB is 10 units and the length of side BC is 6 units. The angle ACB is 90 degrees. Find the length of side AC.



A 10

B 6

C 16

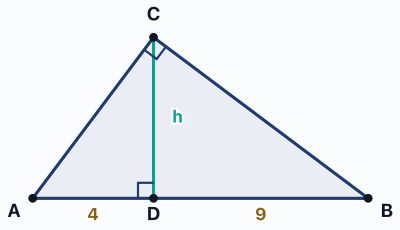
D 8 ✓

WORKED SOLUTION & VERIFICATION

The triangle ABC is a right-angled triangle with the right angle at C. We can use the Pythagorean theorem, which states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. In this case, AB is the hypotenuse. So, we have: $AB^2 = AC^2 + BC^2$. We are given $AB = 10$ and $BC = 6$. We need to find AC. Substituting the given values into the equation: $10^2 = AC^2 + 6^2$ $100 = AC^2 + 36$ $AC^2 = 100 - 36$ $AC^2 = 64$ $AC = \sqrt{64}$ $AC = 8$. Therefore, the length of side AC is 8 units. **[M1] [A1]**

7 MCQ

In the right-angled triangle ABC, shown below, AD is the altitude to the hypotenuse BC. If BD = 4 and DC = 9, find the length of the altitude AD.



A 6 ✓

B 8

C 5

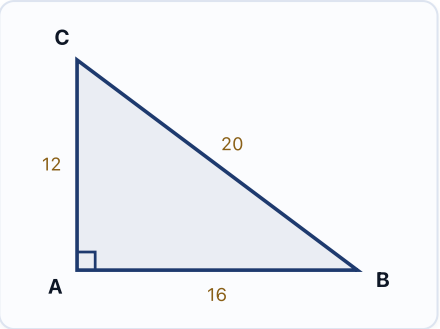
D 7

WORKED SOLUTION & VERIFICATION

In a right-angled triangle, the altitude to the hypotenuse creates similar triangles. By the Geometric Mean Theorem, the square of the altitude to the hypotenuse is equal to the product of the lengths of the two segments it divides the hypotenuse into. Thus, $AD^2 = BD \times DC$. Given $BD = 4$ and $DC = 9$, we have $AD^2 = 4 \times 9 = 36$. Taking the square root of both sides, we get $AD = \sqrt{36} = 6$. **[M1] [A1]**

8 MCQ

A right triangle has legs 12 and 16 and hypotenuse 20. The angle θ is opposite the side of length 12. Find $\cos(\theta)$.



A $\frac{4}{3}$

B $\frac{3}{5}$

C $\frac{4}{5}$ ✓

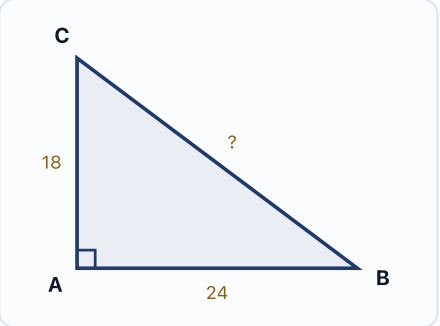
D $\frac{3}{4}$

WORKED SOLUTION & VERIFICATION

$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{16}{20} = \frac{4}{5}$. **[M1] [A1]**

9 MCQ

A right triangle has legs of length 18 and 24. Find the length of the hypotenuse.



A 34

B 33

C 30 ✓

D 32

WORKED SOLUTION & VERIFICATION

By Pythagoras: $18^2 + 24^2 = 30^2$, so the unknown side is 30. **[M1] [A1]**

10 MCQ

Convert 210° to radians, giving your answer as an exact multiple of π.

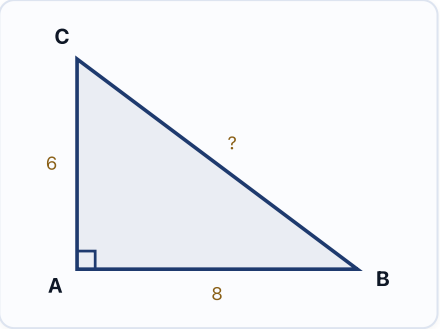
- **A** $\frac{7\pi}{6}$ ✓
- **B** $\frac{5\pi}{3}$
- **C** $\frac{\pi}{2}$
- **D** $\frac{7\pi}{4}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $210^\circ \times \frac{\pi}{180} = \frac{7\pi}{6}$. **[M1] [A1]**

11 MCQ

A right triangle has legs of length 6 and 8. Find the length of the hypotenuse.



- **A** 5
- **B** 10 ✓
- **C** 7
- **D** 0

WORKED SOLUTION & VERIFICATION

By Pythagoras: $6^2 + 8^2 = 10^2$, so the unknown side is 10. **[M1] [A1]**

12 MCQ

Convert 270° to radians, giving your answer as an exact multiple of π.

- **A** $\frac{3\pi}{4}$
- **B** $\frac{\pi}{4}$
- **C** $\frac{3\pi}{2}$ ✓
- **D** $\frac{2\pi}{3}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $270^\circ \times \frac{\pi}{180} = \frac{3\pi}{2}$. **[M1] [A1]**

13 MCQ

Convert $\frac{\pi}{2}$ radians to degrees.

A 30°

B 240°

C 90° ✓

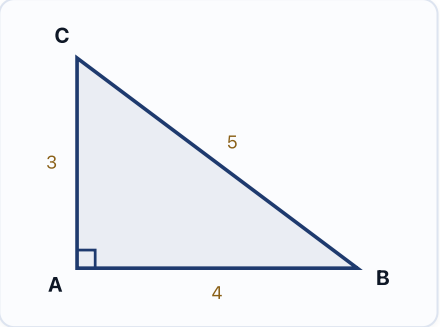
D 315°

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $\frac{\pi}{2} \times \frac{180}{\pi} = 90^\circ$. **[M1] [A1]**

14 MCQ

A right triangle has legs 3 and 4 and hypotenuse 5. The angle θ is opposite the side of length 3. Find $\cos(\theta)$.



A $\frac{3}{4}$

B $\frac{3}{5}$

C $\frac{4}{3}$

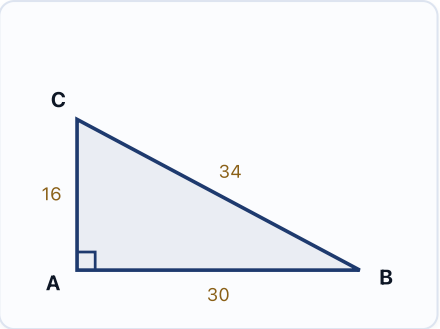
D $\frac{4}{5}$ ✓

WORKED SOLUTION & VERIFICATION

$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{4}{5} = \frac{4}{5}$. **[M1] [A1]**

15 MCQ

A right triangle has legs 16 and 30 and hypotenuse 34. The angle θ is opposite the side of length 16. Find $\sin(\theta)$.



A $\frac{8}{15}$

B $\frac{15}{8}$

C $\frac{15}{17}$

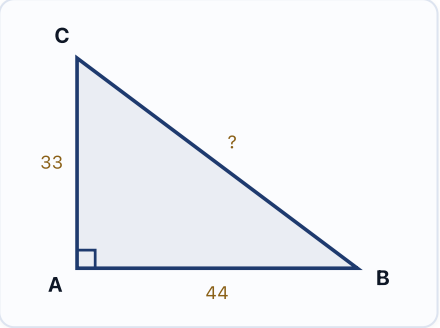
D $\frac{8}{17}$ ✓

WORKED SOLUTION & VERIFICATION

$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{16}{34} = \frac{8}{17}$. **[M1] [A1]**

16 MCQ

In an applied problem, a right triangle has legs of length 33 and 44. Find the length of the hypotenuse.



A 50

B 45

C 65

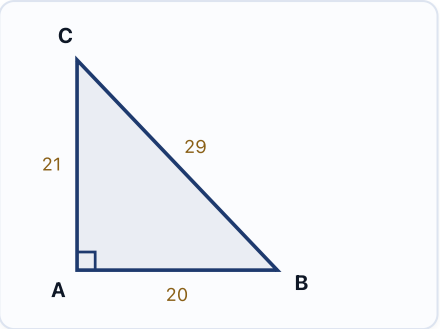
D **55** ✓

WORKED SOLUTION & VERIFICATION

By Pythagoras: $33^2 + 44^2 = 55^2$, so the unknown side is 55. **[M1] [A1]**

17 MCQ

A right triangle has legs 21 and 20 and hypotenuse 29. The angle θ is opposite the side of length 21. Find $\sin(\theta)$.



A $\frac{21}{20}$

B $\frac{20}{21}$

C $\frac{20}{29}$

D $\frac{21}{29}$ ✓

WORKED SOLUTION & VERIFICATION

$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}} = \frac{21}{29} = \frac{21}{29}$. **[M1] [A1]**

18 MCQ

A right triangle has one leg of length 12 and hypotenuse 20. Find the length of the other leg.

A 13

B 6

C 11

D **16** ✓

WORKED SOLUTION & VERIFICATION

By Pythagoras: $12^2 + 16^2 = 20^2$, so the unknown side is 16. **[M1] [A1]**

19 MCQ

A right triangle has one leg of length 9 and hypotenuse 15. Find the length of the other leg.

A **12** ✓

B 8

C 7

D 2

WORKED SOLUTION & VERIFICATION

By Pythagoras: $9^2 + 12^2 = 15^2$, so the unknown side is 12. **[M1] [A1]**

20 MCQ

Convert 240° to radians, giving your answer as an exact multiple of π.

A $\frac{7\pi}{6}$

B π

C $\frac{4\pi}{3}$ ✓

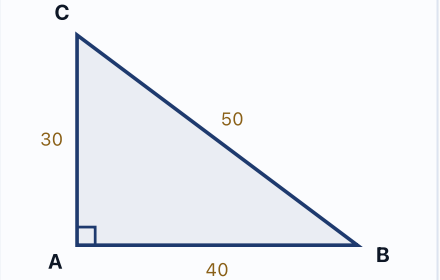
D $\frac{\pi}{3}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $240^\circ \times \frac{\pi}{180} = \frac{4\pi}{3}$. **[M1] [A1]**

21 MCQ

In an applied problem, a right triangle has legs 30 and 40 and hypotenuse 50. The angle θ is opposite the side of length 30. Find $\cos(\theta)$.



A $\frac{4}{3}$

B $\frac{4}{5}$ ✓

C $\frac{3}{4}$

D $\frac{3}{5}$

WORKED SOLUTION & VERIFICATION

$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{40}{50} = \frac{4}{5}$. **[M1] [A1]**

22 MCQ

In an applied problem, a right triangle has one leg of length 9 and hypotenuse 41. Find the length of the other leg.

A 36

B 35

C 37

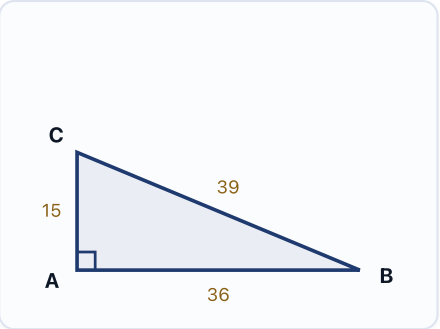
D **40** ✓

WORKED SOLUTION & VERIFICATION

By Pythagoras: $9^2 + 40^2 = 41^2$, so the unknown side is 40. **[M1] [A1]**

23 MCQ

A right triangle has legs 15 and 36 and hypotenuse 39. The angle θ is opposite the side of length 15. Find $\tan(\theta)$.



A $\frac{15}{38}$

B $\frac{5}{12}$ ✓

C $\frac{17}{36}$

D $\frac{12}{5}$

WORKED SOLUTION & VERIFICATION

$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}} = \frac{15}{36} = \frac{5}{12}$. **[M1] [A1]**

24 MCQ

Convert 330° to radians, giving your answer as an exact multiple of π.

A $\frac{11\pi}{6}$ ✓

B $\frac{\pi}{2}$

C $\frac{4\pi}{3}$

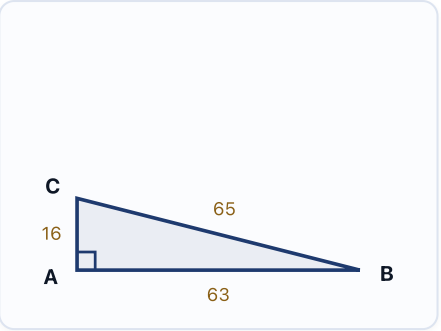
D $\frac{\pi}{3}$

WORKED SOLUTION & VERIFICATION

Working through this trigonometry step: $330^\circ \times \frac{\pi}{180} = \frac{11\pi}{6}$. **[M1] [A1]**

25 MCQ

A right triangle has legs 16 and 63 and hypotenuse 65. The angle θ is opposite the side of length 16. Find $\tan(\theta)$.



A $\frac{16}{63}$ ✓

B $\frac{2}{7}$

C $\frac{63}{16}$

D $\frac{16}{65}$

WORKED SOLUTION & VERIFICATION

$\tan(\theta) = \frac{\text{opposite}}{\text{adjacent}} = \frac{16}{63} = \frac{16}{63}$. **[M1] [A1]**